

ST BENEDICT'S COLLEGE



SUBJECT	Mathematics	DATE	2 March 2017
GRADE	11	MARKS	35
EXAMINER NAME	Mr C Botha	MODERATOR	Mrs Sillman
		DURATION	40 min
CLASS			

QUESTION	ASS STANDARD DESCRIPTION	TOTAL	ACTUAL
1	Exponents Simplify		
2	Exponential Equations		
3	Exponents Application		

NO CALCULATORS ARE ALLOWED ON YOUR DESKS

NO CALCULATORS ARE ALLOWED IN THIS PAPER

Question 1

[19]

Simplify the following

1.1 $\frac{\sqrt{32}}{2} + \sqrt{2}$ (3)

1.2. $\sqrt[3]{8} + \sqrt[3]{27} + \sqrt[3]{-64}$ (2)

1.3. Rationalise the denominator: $\frac{6}{\sqrt{3}}$ (2)

1.4. $\left[\frac{p^{\frac{3}{4}}}{r^{\frac{2}{3}}} \right] \cdot \sqrt[3]{r^{-1}} \cdot p^{\frac{1}{4}}$ (4)

1.5. $\frac{2^{a+4} - 6 \cdot 2^{a+1}}{5 \cdot 2^{a+2}}$ (4)

1.6. $\frac{4^{n+2} \cdot 9^{n-1}}{72^n \cdot 2^{1-n}}$ (4)

Question 2 Solve for x:

[12]

2.1. $3 \cdot 2^x = 48$ (2)

2.2 $9^x + 3^{2x+1} = 108$ (3)

2.3. $2^{2x} + 2^x - 6 = 0$ (4)

2.4. $3(x - 1)^{\frac{1}{2}} = 12$ (3)

[12]

Question 3

Determine the value of: $\frac{2^{2017} + 2^{2019}}{2^{2018}}$ (4)

TOTAL: 35